Black Holes in the W-Structure

Author: Hrachya Danielyan. **Version:** 1.3, November 2026. **License:** CC BY 4.0. Russian original: black_hole_ru.md.

In the W-Structure, a black hole is the limit 3W: all possible links are realized, and nothing distinguishable changes inside. There is no singularity: inside the limit there is no three-dimensional space. Outside, space densifies layer by layer, and the field has the form e^(±δ). With δ = GM/(rc²) this is the exponential metric, the same as in Papapetrou (1954) and Yilmaz (1958). Hence the difference from general relativity (GR): the throat is at 2.72 GM/c², and the shadow is 4.6 % larger (for the unmodified exponential metric, section 4). What is new here is the derivation of the field form from layers and links, and what lies beyond the throat; the numbers for this metric are known.

Labels: [author] — a position of the author; [compendium] — a position of the author's working compendium "W-Structure" v1.31 (unpublished; the positions used are given in section 2); [derived] — follows from what is named; [calculated] — computed from what is named; [condition] — a premise the result depends on; [estimate] — an approximate calculation; [check] — comparison with data or literature.

1. Summary

Statement	Status	Basis
A black hole is the limit 3W: saturation (O3), nothing distinguishable changes inside	author, compendium	O3, positions 1–2
There is no singularity (no infinite quantities)	derived	positions 2, 3, 11; the rate e^(−δ) is finite
The field around: rate e^(−δ), steps per unit path e^(+δ)	derived	O5–O6, positions 7–9
Throat 2.72 GM/c²; shadow 5.44 GM/c² (GR 5.20); ISCO 6.34 GM/c² (GR 6) — for non-rotating layers	calculated under a condition	δ = GM/(rc²), isotropic densification, section 4
Everything that reaches the throat passes into indistinguishability and does not return: the throat itself neither shines nor reflects; there are no echoes and no surface glow	derived	positions 3, 4, 12, 13
Light from matter at the throat arrives with a finite redshift: 1 + z = e for a static emitter, about 12–14 for freely falling matter (depends on the ray direction); in GR it is infinite at the horizon	derived under a condition	sections 4 and 6
The black hole itself (3W) does not rotate; the layers around it rotate, the layer W − 1 not faster than light; rotation of the layers changes the field outside	author, derived	positions 5, 10, 17
The bound χ ≤ 1 is not derived from the Structure	derived	section 5
There is one primordial black hole — at O; there are no black holes lighter than the limit of a non-rotating neutron star (minus mass carried away in collapse)	author, derived under a condition	positions 14, 16
No steady Hawking flux: there is no horizon, and the redshift at the throat is finite	derived under a condition	section 6

2. Basis

Link [author, November 2026]. A link is a stable sequence of acts of change (movements of points). One act is one step and one tick; length is the number of steps. This definition replaces the compendium v1.31 definition (O2: a link is a failed attempt at change); the positions below are given in wordings consistent with it.

Positions of the compendium v1.31:

- **O3** [author]. Closure and saturation are different. A closed system as a whole does not change (conservation laws), but while it is not saturated, new links arise in it. In saturation (W) all possible links are realized: a new act can only go along an existing link and creates no distinguishable change — this is the case of a black hole.
- **O4** [author]. A link that reaches the limit 3W does not disappear; the limit only grows. (In compendium v1.31 this is the general “saturation only grows”; here it is used only for the limit 3W.)
- **O5–O6** [author]. Time is a measure of changes. One step (one act) is the unit of both path and time; the rate of steps is constant; density increases the number of steps along a path.
- **O8–O9** [author]. Mass is the density of local links. (The formulations of mass in the working documents are not yet reduced to one.)
- **G2, G3** [compendium]. Rate e^(−δ); number of steps per unit path e^(+δ).
- **G7** [compendium]. A black hole is the limit 3W, without a singularity.
- **G8** [compendium]. No Hawking radiation, no black holes lighter than the maximum neutron-star mass, no echoes.
- **G9** [compendium]. “Rotation” is a property of the horizon and the medium; a ceiling from nested rings; $\chi > 1$ possible (estimate). The G9 estimate is not adopted here: the ceiling on χ is not derived (section 5). The “horizon” in G9 is the throat layer (section 3).
- **G1** [compendium: proposal]. Two channels (presence and motion) → light deflection 1.75″.
- **G4** [compendium: calculation; relies on G1 and G2]. Gravity Probe B: geodetic precession ≈ 6606 mas/yr (measured 6602 ± 18), frame dragging ≈ 39.2 mas/yr (37.2 ± 7.2).
- **F13** [compendium; open]. 3W = 40 or 00.

Positions of the author [author]:

1. 3W — all possible links are realized, everything “freezes”. Inside 3W there are no distinguishable changes; the boundary grows by what arrives from outside. 3W is the same saturation as in O3.
2. “Infinite approach” to the limit is in fact a difference of one discrete step.
3. The throat is the smallest circle, where growth inward ends; further inward there is no three-dimensional space.
4. A black hole grows one-dimensionally: infalling matter does not hit a wall but merges with “nothing” (from the point of view of three-dimensional space).
5. One cannot speak of the rotation of the black hole itself: 3W does not rotate. The layer $W - 1$ rotates; the limit W itself only grows. Matter in the outer layers orbits at different speeds, as in any gravity.
6. There is no mass in the “core”. Mass is in the layers around it; the closer a layer is to the limit, the greater its mass.
7. Densification of path and slowing of time are one phenomenon: more steps for the same path.
8. The number of steps per unit path changes from layer to layer by the same factor as the rate.
9. The distance r is counted in layers: there is one unit of r between neighbouring layers, while the number of steps per layer grows.
10. One indivisible change takes one indivisible interval of local time.
11. Radiation from 3W is impossible: there is absolute self-consistency there — indistinguishability; there is no three-dimensional space.
12. Energy and mass that reach the throat turn into indistinguishability “inside” and increase the diameter of the throat.
13. (Consequence of 3, 4, 12.) Nothing that has reached the throat returns to three-dimensional space.
14. There is one primordial black hole — at the zero point O. All others are not primordial: away from O, only collapse leads to 3W.
15. Matter in the layer $W - 1$ is held back: one step remains to the limit, and the inflow is one-dimensional, so it does not enter in one step and orbits before merging with “nothing”. The direction and magnitude of this rotation are set by the angular momentum of the progenitor star: it is conserved in contraction (O3), and the layers spin up. All real stars rotate, so the layers practically always rotate; with strictly spherical infall without rotation there would be no reason to rotate.
16. [condition] Collapse away from O is the collapse of matter that a neutron star cannot hold; there is no other path to 3W (this condition also excludes light black holes from early densifications; it does not follow from position 14: their formation is also a collapse).
17. Rotation of the layers affects the field outside (the motion channel, as in G1 and G4). Therefore the shadow may also depend on the rotation of the layers.
18. At the throat, self-closed links do not break (O4: a link does not disappear) but lose distinguishability. A form is a distinguishable group of links; when all possible links are realized, there is nothing to single a group out

from. This is the mechanism of position 12.

19. The mechanism is the same for the point O and for all other black holes: realization of all possible links. The point O became a black hole this way; the others reach it through collapse (positions 14, 16).

3. What a black hole is

3W is saturation [author, O3]. All possible links are realized. Every new act goes along an already existing link and creates no distinguishable change. Inside there are no distinguishable changes; the boundary grows by what arrives from outside (position 1).

There is no singularity [derived]. A singularity here is a place where quantities become infinite. There is no such place:

- inside the limit there is no three-dimensional space (position 11) — so there is no point where quantities could go to infinity;
- outside, the rate $e^{-\delta}$ is finite; at the throat $\delta = 1$ (section 4), and the rate is $e^{-1} \approx 0.37$ of the distant one;
- one discrete step remains to the limit (position 2).

Paths of bodies end at the throat: what reaches it passes into indistinguishability and increases the throat (position 12). This is not a break in the description but a transition into 3W.

There is no event horizon in the sense of GR. In GR the rate drops to zero at the horizon. Here the rate at the throat is e^{-1}: there is no surface of infinite slowing. Below, “throat layer” is used instead of “horizon”.

What lies beyond 3W [compendium F13; open]. The transition is 3W = 40 (the next dimensionality) or 3W = 00. The choice depends on whether it is proved that three dimensions are the limit of W for dimensionality.

4. Space around

Rate [derived from O5–O6; G2]. The rate is defined only relative to the neighbouring layer, so the rate ratios of neighbouring layers multiply. If the logarithm of the ratio of neighbouring layers equals the increment of δ, and the step between layers is small compared with the scale on which δ changes, the product gives $e^{-\delta}$ relative to a distant observer.

Densification of path [derived from positions 7–8]. The number of steps per unit path changes from layer to layer by a factor of the same size, inverse to the rate factor, so there are $e^{+\delta}$ steps per unit r. This is the same phenomenon as the slowing of time. The factor is the same along the radius and around the circle [condition: isotropic densification]; without densification the circumference of layer r is $2\pi r$.

The magnitude of δ [condition]. $\delta = GM/(rc^2)$, where r is the layer count (position 9). Checks of this law:

- the perihelion of Mercury is a second-order check in $GM/(rc^2)$. Both factors are needed: the second-order term in the rate and the first-order path factor $e^{+\delta}$. With $e^{\pm\delta}$ the result is the same as in GR — about 43″ per century; the observed excess agrees. The rate alone, without the path factor, would give only a third, about 14.3″;
- the perihelion also fixes how the layer factors combine [calculated]. The factors multiply (O5–O6), so the rate is $e^{-\delta}$ and $g_{00} = -(1 - 2\delta + 2\delta^2 - \ldots)$. If they combined additively instead — rate $1/(1 + \delta)$, path factor $1 + \delta$, with the same first order — the second-order term would be $3\delta^2$ instead of $2\delta^2$ (PPN $\beta = 3/2$ instead of 1), and the perihelion would be about 35.8″ per century instead of 43″. So the observed perihelion supports the multiplicative composition, not only the first-order 1/r law;
- light deflection at the Sun with $e^{\pm\delta}$ is 1.75″, as in G1. The “two channels” of G1 are exactly the rate and the path factor $e^{+\delta}$, not an addition to them; otherwise the contribution would be counted twice.

Near the throat, where $\delta \sim 1$, the law is not tested (section 9). The circular photon orbit lies at $\delta = 1/2$, inside this untested region. The perihelion and light deflection fix the law only up to second order in δ. A third-order change leaves them unchanged but shifts the shadow: for example, a rate $e^{-\delta+\varepsilon\delta^3}$ changes the critical impact parameter by $-\varepsilon/8$ in relative terms. So the shadow numbers below are predictions of the unmodified exponential metric, not of every law with the same weak field.

Where the mass is and under which condition the numbers hold [condition]. The numbers below are obtained for $\delta = GM/(rc^2)$, that is, as if all of the mass M lies deeper than the layer considered. By position 6 the

mass lies in the layers around, denser closer to the limit. So the numbers hold outside the massive layers. If the massive layers are thin compared with GM/c^2, the numbers hold approximately near the throat as well. The law of mass distribution over the layers is not derived (section 9).

Throat [calculated under a condition]. The circumference of layer r is $2\pi \cdot r \cdot e^{(GM/(rc^2))}$. It is smallest at $r = GM/c^2$ ($\delta = 1$); its circumferential radius is $e \cdot GM/c^2 \approx 2.72\ GM/c^2$.

Observable numbers [calculated under a condition]. Radii are circumference divided by 2π, in units of GM/c^2.

Quantity	Structure	GR (non-rotating)	Difference
Throat (GR: horizon)	$e \approx 2.72$	2	— (not observed directly)
Circular photon orbit	$2\sqrt{e} \approx 3.30$	3	+9.9 %
Shadow radius (critical impact parameter)	$2e \approx 5.44$	$3\sqrt{3} \approx 5.20$	+4.6 %
ISCO (innermost stable circular orbit)	6.34	6	+5.6 %

Thin bright subrings in the image converge to the critical curve of diameter $4e \approx 10.87$ (GR $6\sqrt{3} \approx 10.39$). This is the same quantity as the shadow — one check, but independent of the model of the emitting gas.

Relation to known results [check]. A field with rate $e^{(-\delta)}$ and path factor $e^{(+\delta)}$ at $\delta = GM/(rc^2)$ is the exponential metric of Papapetrou (1954) and Yilmaz (1958). The same numbers have been obtained for it in the literature: a throat with circumferential radius $e \cdot GM/c^2$, a circular photon orbit $2\sqrt{e} \cdot GM/c^2$, ISCO $\approx 6.338\ GM/c^2$; the shadow radius $2e \cdot GM/c^2$ follows from the same work (Boonserm, Ngampitipan, Simpson, Visser, 2018). The agreement is a check of the calculation, not a priority claim.

What is our own here:

- the derivation of the field form: rate and link density change from layer to layer by one factor (positions 7–8), r is the layer count (position 9);
- what lies beyond the throat. In the literature the throat of the exponential metric is a traversable wormhole into another region of space. In the Structure there is no three-dimensional space beyond the throat: there is 3W, and what reaches it does not return (positions 3, 12, 13).

Growth [derived]. The throat $e \cdot GM/c^2$ is proportional to the mass. By position 12 everything that reaches the throat increases it; by O4 the limit only grows.

5. Rotation

What rotates [author, position 5]. The black hole itself (3W) does not rotate. The layer $W - 1$ rotates; the limit W itself only grows. Matter in the outer layers orbits at different speeds; this is a property of any gravity and by itself does not distinguish the Structure from GR.

Why there is rotation [author, position 15; O3]. A black hole forms from a rotating star. Angular momentum is conserved in contraction (O3), so the layers spin up: the smaller the radius, the faster the revolution. In the layer $W - 1$ matter is held back — one step remains to the limit, and the inflow is one-dimensional — and orbits before merging with “nothing”. Strictly zero rotation would require a strictly non-rotating star; real stars are never like that.

Speed limit [derived from position 10 and O6]. One indivisible change takes one indivisible interval of local time. Hence the layer $W - 1$ does not move faster than light in local time.

The bound $\chi \leq 1$ is not derived from the Structure [derived]. $\chi = Jc/(GM^2)$, where J is the total angular momentum of the layers (3W itself does not rotate, position 5) and M is the mass seen from outside; it is not a speed.

- In GR $\chi \leq 1$ because at $\chi > 1$ the horizon disappears and a singularity is exposed. For an extremal rotating GR black hole the equatorial circumferential radius of the horizon is $2GM/c^2$.
- In the Structure there is no singularity, and the throat is larger: $e \cdot GM/c^2$ versus $2GM/c^2$, $e/2 \approx 1.36$ times.
- The GR argument for $\chi \leq 1$ relies on a singularity that does not exist here. The Structure’s own limit on χ is not derived: a simple Newtonian estimate (a ring at the throat moving at the speed of light) proves nothing — for an extremal Kerr black hole it would give $\chi = 2$ instead of the true 1.

Rotation of the layers changes the field [author, position 17]. Motion of the source mass affects the field: the compendium calculates the frame dragging of the Gravity Probe B gyroscopes this way (G4; relies on G1 [proposal] and G2). This is not the same motion channel of G1 as in section 4: there the light moves, not the source; how G4 is obtained from G1 is not given in the compendium. Hence [derived under a condition]: rotating layers drag their surroundings, and the field at the throat, and with it the shadow and the ISCO, may depend on the rotation of the layers. The law of this influence near the throat is not derived; the ceiling on χ is not computed (section 9).

6. What is seen from outside

What reaches the throat does not return [derived from positions 3, 4, 12]. Matter, light and waves that reach the throat pass into the indistinguishability of 3W and increase the throat. Hence:

- **the throat itself neither shines nor reflects**: rays with impact parameter less than $2e \cdot GM/c^2$ reach the throat and do not return; inside the shadow only the light of gas lying on the line of sight in front of the throat is seen; light from gas at the throat itself is strongly dimmed;
- **there are no echoes**: an echo in merger signals would arise from waves reflecting off a surface, and there is nothing to reflect;
- **there is no surface glow**: there is nothing to heat up; the energy of infalling matter passes into indistinguishability.

Last light [derived under the condition of section 4]. There is no event horizon, so an infalling body reaches the throat in a finite time of the distant observer. Light emitted at the throat itself arrives with a finite redshift: $1 + z = e \approx 2.72$ for a static emitter; for matter freely falling from rest at infinity, from ≈ 12 to ≈ 14 depending on the ray direction (≈ 14 for a ray strictly outward; rays deviating from the radius by more than $\approx 47°$ are captured); dimming $2 \cdot 10^4$–$4 \cdot 10^4$. In GR the redshift at the horizon is infinite. This difference is observable in principle; the brightness inside the shadow has to be calculated (section 9).

No steady Hawking flux [derived under a condition]. The thermal Hawking flux does not need matter inside; it arises in the quantum fields outside, and it requires an exponential stretching of light rays — a redshift that grows without bound (a horizon is not required, but this stretching is; Barceló, Liberati, Sonego, Visser, 2011). The exponential metric has no horizon, and the redshift at the throat is finite (section 6, “Last light”). Therefore the standard mechanism gives no steady thermal flux. This does not say that Hawking’s calculation is wrong; it says that it does not apply to this throat. Open: a transient burst while the geometry changes during formation, and the effect of the throat as a perfectly absorbing boundary on quantum fields (section 9). Separately, by position 11 nothing radiates from 3W itself [author].

7. Masses of black holes

There is one primordial black hole [author, position 14]. It is the black hole at the zero point O. All observed black holes are not primordial: away from O, only collapse leads to 3W.

No black holes lighter than the neutron-star limit minus the mass carried away [derived under a condition, positions 14, 16]. Collapse reaches 3W only when a neutron star cannot hold the matter. Hence the mass of any observed black hole is not less than the limit of a non-rotating neutron star minus the mass carried away in collapse. (Rotation raises the limit for the star itself by about 20 %.) The value of the limit is not computed in the Structure. The heaviest measured neutron stars are about 2.1 and 2.35 solar masses (PSR J0740+6620 and PSR J0952−0607).

8. Predictions and tests

Prediction	Structure	GR	How to test	Now
Shadow size (and the critical curve of the subrings) for non-rotating layers	radius $2e \approx 5.44\ GM/c^2$, 4.6 % larger	$3\sqrt{3} \approx 5.20\ GM/c^2$	EHT and future interferometry with a space baseline; the mass of Sgr A* from stellar orbits	Sgr A*: deviation from GR -0.08 ± 0.09 (VLTI mass) and -0.04 ($+0.09/-0.10$) (Keck); the Structure gives $+0.046$ — at 1.4σ and 1.0σ. Not excluded. Distinguishing requires 2–3 % accuracy in size

				and mass together. EHT converts ring size to shadow size using GR models
Light from matter at the throat	finite redshift (about 12–14 for free fall)	redshift at the horizon is infinite	image inside the shadow, flares at the edge	brightness calculation needed
No black holes lighter than the neutron-star limit (minus mass carried away in collapse)	—	allows light primordial ones	a securely established black hole lighter than the heaviest securely measured neutron star (now about 2.1 solar masses, PSR J0740+6620) would refute the prediction, if little mass was carried away at its formation	none known. Objects in the “mass gap” (GW230529: 2.5–4.5; GW190814: about 2.6) are heavier than the measured neutron stars and do not contradict the prediction: if they are black holes, the limit is not above their mass
No echoes	—	no echoes in GR either	echo searches after mergers	not found; consistent, but does not distinguish from GR
No steady Hawking flux (no horizon, finite redshift; a burst during formation is open, section 9)	—	present, but immeasurably small for stellar black holes	only for light black holes, which do not exist according to the Structure	practically untestable
ISCO 6.34 GM/c²	—	6 GM/c² without rotation	disk edge, oscillations	cannot be distinguished without the field law for rotating layers: a Kerr black hole with $\chi \approx -0.10$ gives the same value

The main test is the shadow size: a 4.6 % difference for non-rotating layers and the unmodified exponential law at $\delta \approx 1/2$ (section 4). By position 15 the layers of real black holes rotate (angular momentum of the progenitor star); rotation of the layers, like rotation of the black hole in GR, may change the size and shape of the shadow, and the law for this is not derived (9.5, 9.11). Until it is derived, the shadow size is not a single number free of fitting. Spin estimates are currently obtained in Kerr models with the bound $\chi \leq 1$; testing χ requires signal models based on the Structure.

9. Open questions

1. **3W = 40 or 00.** Depends on a proof that three dimensions are the limit of W for dimensionality (F13).
2. **Mass distribution over the layers and δ(r).** The numbers of section 4 hold outside the massive layers. A law of mass distribution over the layers (position 6) and δ(r) inside the massive layers are needed. Related: which links count as mass. If, following O8, the densification of links e^(+δ) around the black hole itself counted as mass, its total mass would grow without bound with distance; so something else must count as mass (for example, its change from layer to layer). In the author’s working document on the mechanism of gravity (November 2026, unpublished) this is settled by a position of the author [author]: mass is the density of self-closed links; links that continue into the medium are not mass. Then the densification e^(+δ) is the state of the medium around the mass, not mass, and is not counted in M. Still open: how the mass M seen from outside is distributed over the layers (position 6) and how it grows (question 9). The other formulations of mass in the working documents are not yet reduced to this one.
3. **The law of δ near the throat.** It is checked against the perihelion to second order. At $\delta \sim 1$ it is not derived from the structure. The shadow depends on it: the photon orbit lies at $\delta = 1/2$ (section 4).
4. **Does the layer W − 1 radiate?** Position 11 closes 3W. The layer W − 1 is not saturated and rotates; there are changes in it. A reason why it does not radiate is needed.
5. **The field of rotating layers.** A law is needed by which the rotation of the layers (frame dragging, G4) changes the field at the throat. The size and shape of the shadow, the ISCO, the ceiling on χ and the difference from GR for rotating systems depend on it.
6. **Field equations and merger signals.** Without them there is no prediction of the ringdown after a merger. In the Structure waves do not return from the throat; the spectrum of these oscillations has to be calculated.
7. **Brightness inside the shadow.** A calculation of the emission of gas on rays inside the shadow is needed, including between the photon orbit and the throat, taking the motion of the gas into account.

8. **The maximum neutron-star mass in the Structure.** At present the bound is taken from observations.
9. **How the external mass M grows.** There is no mass in the core (position 6), and what reaches the throat becomes indistinguishability and increases the throat (position 12). How the mass M seen from outside grows, and how it relates to the mass of the layers, is not derived.
10. **The black hole at the point O and the edge of space** (F5, F11). Not considered in this file.
11. **The shadow for rotating layers.** The layers of real black holes rotate (position 15: angular momentum of the progenitor star). The numbers of section 4 and the shadow 2e are given for non-rotating layers — the limit of slow rotation. A law is needed by which the rotation of the layers changes the size and shape of the shadow (see 9.5); until then the comparison with observations is fair only for black holes with slowly rotating layers.
12. **The path of collapse (condition 16).** Why, away from O, there is no path to 3W that bypasses a neutron star (for example, early densifications). How much mass can be carried away in collapse: without this bound the threshold of section 8 is not defined.
13. **Entropy and area.** Inside 3W there are no distinctions (position 11). Therefore everything by which one black hole differs from another lies on the throat, and the number of its states is set by the area of the throat, not by its volume. This agrees with the Bekenstein–Hawking law (entropy ∝ area). Not derived: (a) the link "number of throat points → number of states"; (b) the numerical factor (in the standard formula, 1/4 in Planck units): the step length is not defined in the Structure, so the numbers cannot be compared.
14. **Emission during formation and the absorbing boundary.** The steady Hawking flux is absent (section 6). Not calculated: a transient burst while the geometry changes during collapse, and what the throat, as a boundary that absorbs everything, does to quantum fields.

Sources

- Danielyan H. "W-Structure": working compendium of positions v1.31 and register of statements (2026), unpublished. The positions this file relies on are given in section 2.
- Papapetrou A. Eine Theorie des Gravitationsfeldes mit einer Feldfunktion. Zeitschrift für Physik 139, 518 (1954).
- Yilmaz H. New approach to general relativity. Physical Review 111, 1417–1426 (1958).
- Boonserm P., Ngampitipan T., Simpson A., Visser M. The exponential metric represents a traversable wormhole. Physical Review D 98, 084048 (2018).
- Event Horizon Telescope Collaboration. First Sagittarius A* EHT results. VI. Testing the black hole metric. Astrophysical Journal Letters 930, L17 (2022).
- LIGO, Virgo, KAGRA. Observation of gravitational waves from the coalescence of a 2.5–4.5 M☉ compact object and a neutron star (GW230529). Astrophysical Journal Letters (2024), doi:10.3847/2041-8213/ad5beb.
- LIGO, Virgo, KAGRA. Tests of general relativity with GWTC-3. Physical Review D 112, 084080 (2025), doi:10.1103/PhysRevD.112.084080.
- Romani R. W. et al. PSR J0952−0607: the fastest and heaviest known Galactic neutron star. Astrophysical Journal Letters (2022), doi:10.3847/2041-8213/ac8007.
- Hawking S. W. Particle creation by black holes. Communications in Mathematical Physics 43, 199–220 (1975).
- Barceló C., Liberati S., Sonego S., Visser M. Hawking-like radiation from evolving black holes and compact horizonless objects. Journal of High Energy Physics 02 (2011) 003.
- Fonseca E. et al. Refined mass and geometric measurements of the high-mass PSR J0740+6620. Astrophysical Journal Letters 915, L12 (2021): 2.08 ± 0.07 solar masses.